



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to AMMAN PETROLEUM

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

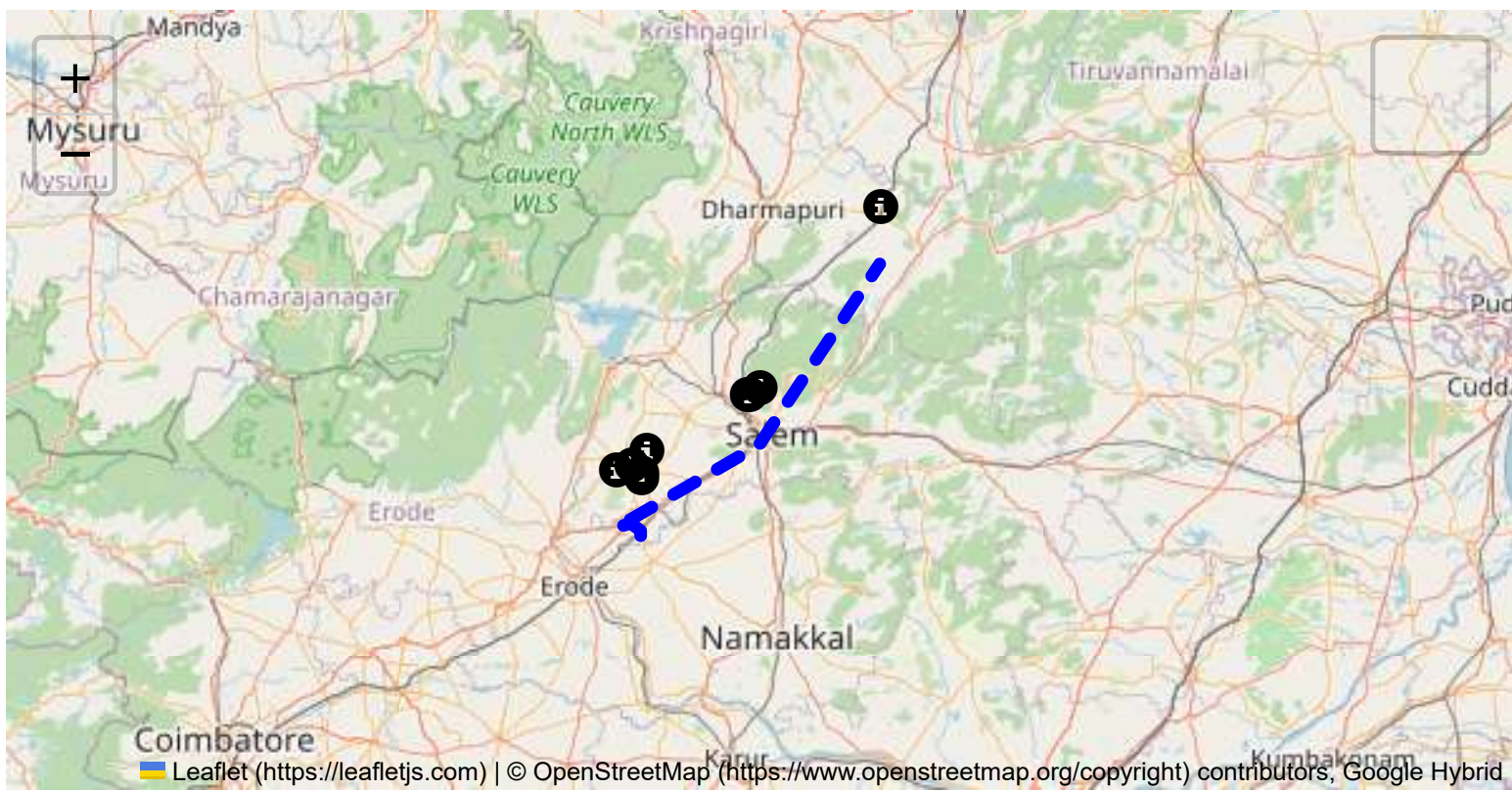
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 108.32 km
Estimated Duration: 2.6 hours
Adjusted Duration (Heavy Vehicle): 3.2 hours
Start: (11.4381, 77.8734)
End: (12.01645, 78.3948)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to Thiruvalluvarnagar, Pappireddipatti spans approximately 108.32 kilometers and passes through various regions including Sankari, Salem, and Neykkarapatti. The primary route involves NH544 and Bhavani Main Road, with sections passing through urban and semi-urban areas.

2. Typical Weather Conditions and Potential Weather-Related Hazards

The typical weather in Tamil Nadu includes a hot climate with temperatures ranging from 25°C to 35°C. The monsoon season brings heavy rainfall from June to September, which can lead to waterlogged roads and landslides in certain areas, particularly on highways. Drivers should be cautious of reduced visibility and slippery roads during the monsoon.

3. Analysis of Traffic Patterns

- **Peak Hours:** Morning (8:00 AM - 10:00 AM) and evening (5:00 PM - 8:00 PM) see major congestion, particularly near urban centers like Salem.
- **Congestion-Prone Areas:** NH544, especially near Salem, and the Sankari RS - Erode Privu Road.
- **Suggestions:** Travel during non-peak hours is advisable to avoid significant delays.

4. Assessment of Road Quality and Infrastructure

- **NH544:** Generally good quality, but prone to bottlenecks near Salem.
- **Bhavani Main Road:** Varies in quality with some sections requiring careful navigation due to potholes.
- **Urban Areas:** Streets can be narrow with occasional encroachments in Salem and Sankari.

5. Suggestions for Alternative Routes for Emergencies

Alternative routes via smaller state highways parallel to NH544 can be used in case of major disruptions. Specific alternatives would require real-time traffic data for optimal selection.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu has specific regulations for the transport of hazardous materials, including permits for travel through certain zones and time restrictions to reduce traffic impact. Drivers must ensure all necessary paperwork is in order and adhere to prescribed routes and timings.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

This region has experienced incidents primarily due to overloading and speeding. Specific black spots, especially near Salem, have recorded accidents involving heavy vehicles. Proper load management and adherence to speed limits can mitigate risks.

8. Environmental Considerations and Sensitive Areas

The route traverses through agricultural regions and some semi-urban settlements. Avoiding accidental spills is crucial to protect local water bodies and farmlands. Enhanced vigilance near populated and environmentally sensitive zones is essential.

9. Analysis of Communication Coverage

Mobile network coverage is generally strong in urban areas but may exhibit some dead zones in rural stretches, particularly between Sankari and Salem. Planning is essential to ensure communication devices are capable of operating over multiple networks.

10. Estimated Emergency Response Times for Different Route Segments

- **Urban (e.g., Salem):** 15-20 minutes due to proximity to emergency services.
- **Rural:** 30-60 minutes, depending on location. Emergency services may take longer to reach remote areas.

12. Overall Summary of Risk Assessment

This route presents moderate risks primarily related to traffic congestion and weather-related disruptions. The infrastructure is adequate but needs monitoring for potholes and bottlenecks. Emergency plans should include alternative routes and network reliability checks for communication. Adherence to safety regulations and cautious driving will minimize incident risks, particularly with hazardous materials.

Overall, this is a viable route for transporting hazardous materials, given that all safety protocols are diligently followed.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63266, 78.12699	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	Medium	11.44900, 77.87406	30 KM/Hr
4	Turn	High	11.45352, 77.85698	15 KM/Hr
5	Turn	Medium	11.45847, 77.85140	30 KM/Hr
6	Turn	Medium	11.45898, 77.85135	30 KM/Hr
7	Turn	High	11.46303, 77.84945	15 KM/Hr
8	Turn	Medium	11.45571, 77.81702	30 KM/Hr
9	Turn	High	11.45615, 77.81677	15 KM/Hr
10	Turn	Medium	11.53552, 77.95763	30 KM/Hr
11	Turn	High	11.61233, 78.10397	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
12	Turn	High	11.61216, 78.10423	15 KM/Hr
13	Turn	High	11.63300, 78.12746	15 KM/Hr
14	Turn	Medium	11.65825, 78.20469	30 KM/Hr
15	Turn	Medium	11.66366, 78.20649	30 KM/Hr
16	Turn	Medium	11.67129, 78.22978	30 KM/Hr
17	Turn	High	11.67120, 78.23136	15 KM/Hr
18	Turn	High	11.67182, 78.23141	15 KM/Hr
19	Turn	Medium	11.67422, 78.23881	30 KM/Hr
20	Turn	Medium	11.68017, 78.24250	30 KM/Hr
21	Turn	Medium	11.83514, 78.35305	30 KM/Hr
22	Turn	High	11.93905, 78.40417	15 KM/Hr
23	Turn	Medium	11.93960, 78.40395	30 KM/Hr
24	Turn	Medium	11.94039, 78.40200	30 KM/Hr
25	Turn	Medium	11.95570, 78.39975	30 KM/Hr
26	Turn	Medium	11.98751, 78.39991	30 KM/Hr
27	Turn	Medium	11.98876, 78.39941	30 KM/Hr
28	Turn	High	12.01020, 78.39698	15 KM/Hr
29	Turn	Medium	12.01049, 78.39755	30 KM/Hr
30	Turn	Medium	12.01222, 78.39878	30 KM/Hr
31	Turn	Medium	12.01366, 78.39888	30 KM/Hr
32	Turn	Medium	12.01431, 78.39858	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
4	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
5	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
8	hospital	Vinayaka Mission's Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
12	hospital	Thayammal Hospitals	11.6237472, 78.1177631	30 km/h	Medium
13	hospital	Vanitha Hospital	11.6239931, 78.1184622	30 km/h	Medium
15	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
16	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
17	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
18	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
19	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
20	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
21	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
22	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
23	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
24	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
25	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
28	hospital	M. Thathanur Primary Health Center	11.6824017, 78.2445813	30 km/h	Medium
29	hospital	Goverment Hospital	11.7580777, 78.2969128	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
6	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
9	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
10	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
11	school	Sri Vidhya Bharati Matric HR Sec School	11.6186069, 78.1103239	30 km/h	Medium
14	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
26	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
27	school	MettupattiThathanur Primary School	11.6827707, 78.2442509	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63266, 78.12699



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45847, 77.85140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45898, 77.85135



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45571, 77.81702



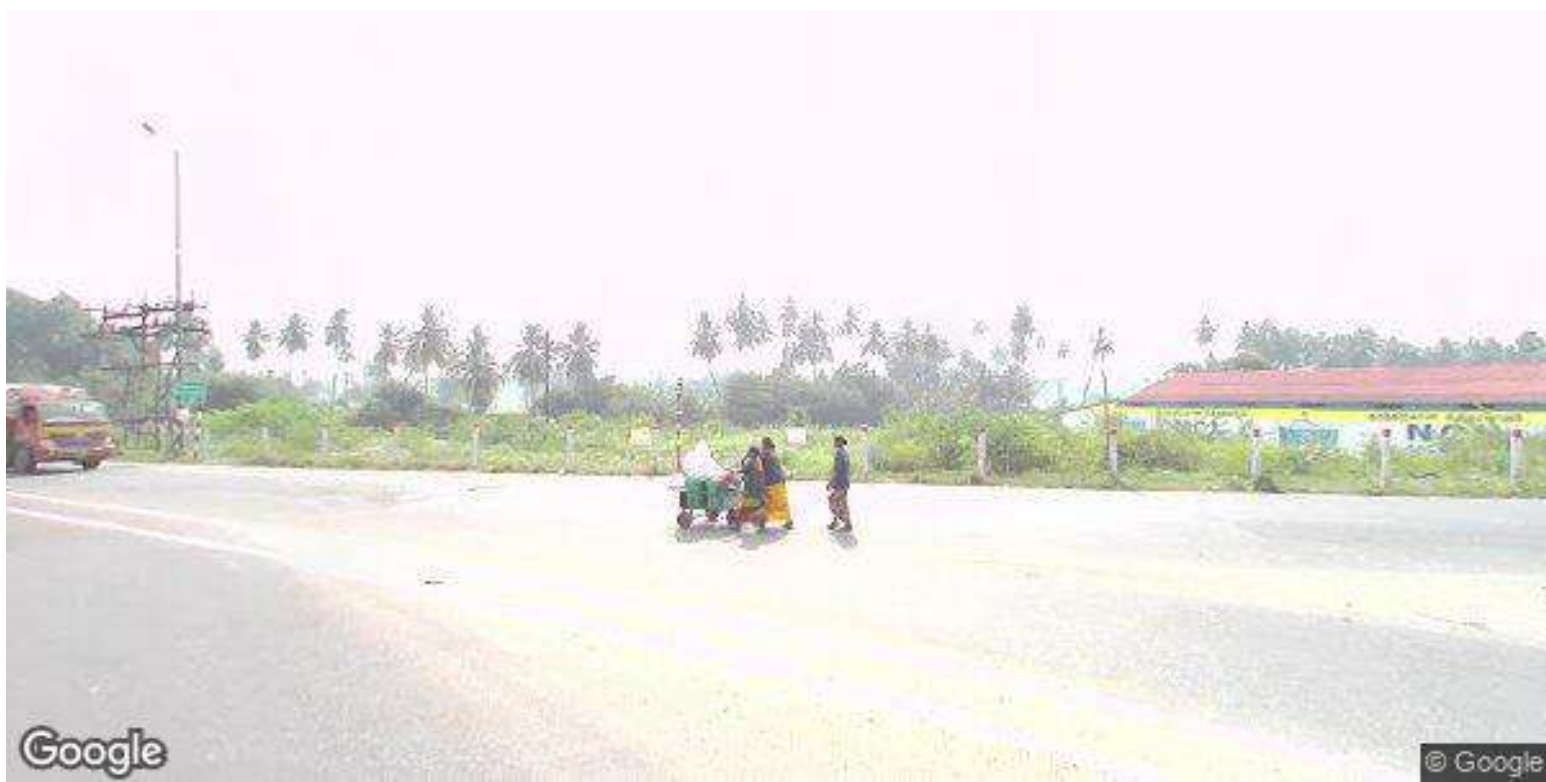
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45615, 77.81677



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53552, 77.95763



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.61233, 78.10397



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.61216, 78.10423



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63300, 78.12746



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65825, 78.20469



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66366, 78.20649



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67129, 78.22978



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67120, 78.23136



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67182, 78.23141



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68017, 78.24250



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.83514, 78.35305



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.93905, 78.40417



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93960, 78.40395



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.94039, 78.40200



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95570, 78.39975



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.98751, 78.39991



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.98876, 78.39941



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.01020, 78.39698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.01049, 78.39755



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.01222, 78.39878



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.01366, 78.39888



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.01431, 78.39858
